

# Four Wheeler Renewal Portal

## Investigation of a Customer-Facing Renewal Flow Failure

### Issue Summary

During the vehicle insurance renewal process, the portal becomes unusable due to multiple UI and JavaScript failures.

- Renewal process cannot be completed
- Blank popup dialog appears
- Multiple overlapping modal windows rendered
- Clicking OK does not allow the user to proceed

# Environment & Reproducibility

## Desktop Browsers

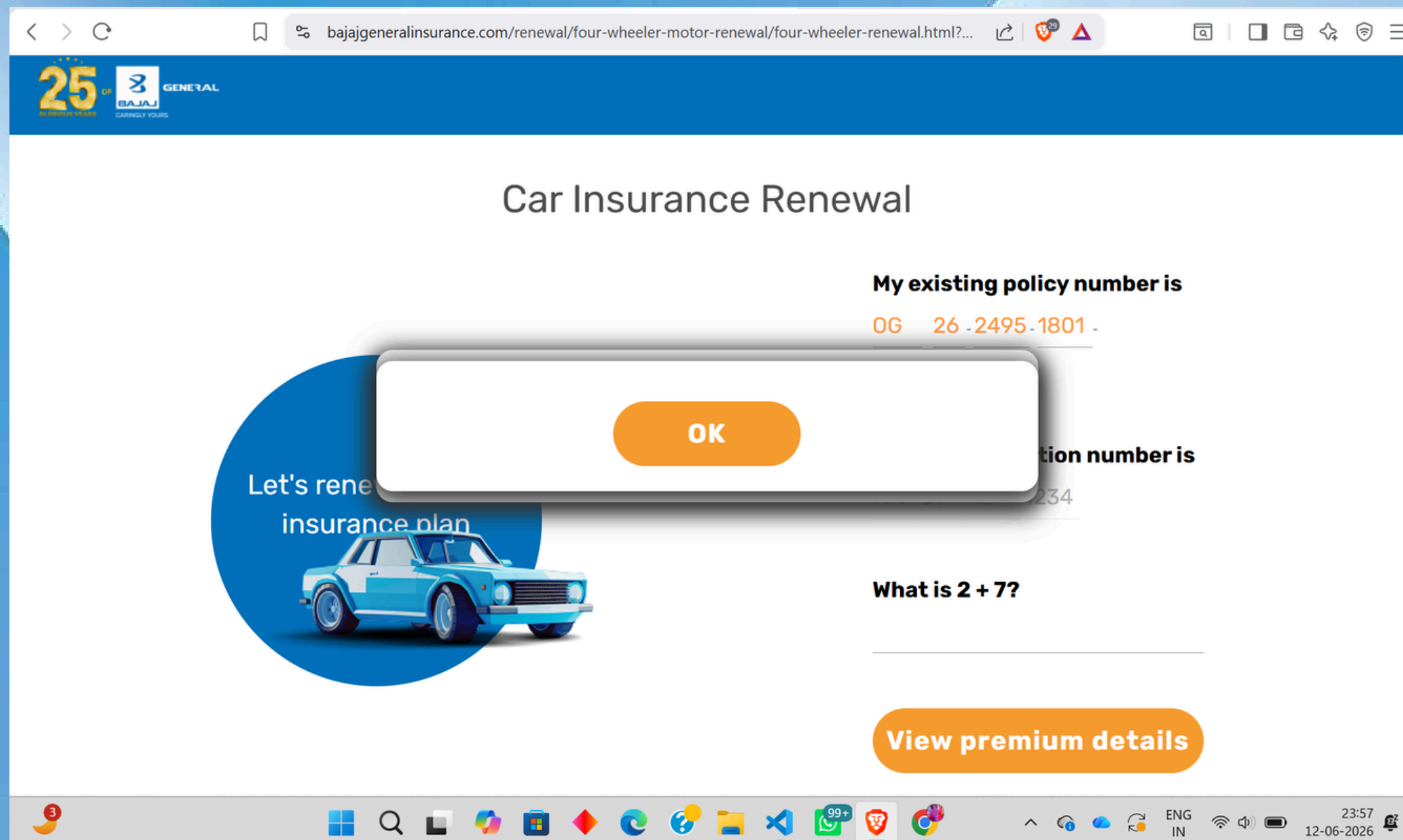
- Google Chrome
- Microsoft Edge
- Brave Browser

## Mobile Browsers

- Chrome Mobile
- Safari

Issue was reproducible across multiple browsers and devices, reducing the likelihood of a client-side environment issue

# Visual Evidence



## Observed Behavior

- Two modal windows visible
- Missing dialog content
- Browser dependent positioning
- User interaction blocked

## Overlapping Dialogs

- Primary modal in foreground
- Secondary modal behind foreground
- Modal body content missing
- Only OK button visible

These observations suggest a modal lifecycle or error handling issue, but source code access would be required for confirmation.

# Console Analysis

## Critical Runtime Error #1

**Uncaught ReferenceError: smartech is not defined**

Location: four-wheeler-renewal.html:450

Meaning: The application attempts to invoke the Smartech SDK before ensuring the library is loaded.  
Consequences: event handling interruption, incomplete workflow execution, unhandled exceptions.

## Critical Runtime Error #2

**Uncaught (in promise) TypeError: Cannot read properties of undefined (reading '0')**

Location: clientlibs...js:2781

Meaning : Application attempts to access an array/object element that does not exist.

Note: Error relationships and potential causes are inferred from observed browser behavior and console output. Confirmation would require source code and backend log access.

# Hypothesis #1

## Evidence

smartech is not defined

### Probable Scenario

Expected:

```
smartech.track(...)
```

Actual:

```
smartech === undefined
```

## Possible Explanation

The application assumes a third party library is loaded without validating its availability at runtime.

## Recommended Fix

Before calling `smartech.track(...)`, validate:

```
if (window.smartech) {  
    smartech.track(...);  
}
```



# Hypothesis #2

## Evidence

Cannot read properties  
of undefined (reading  
'0')

## Probable Scenario

Expected: { "data": [...] }  
}

Actual: { } or { "data":  
null }

`response.data[0]` →  
immediate failure

## Recommended Fix

Implement defensive  
validation:

```
if  
(response?.data?.length) {  
  // safe to access  
}
```

# Third-Party Dependencies Observed

## Adobe DTM

`AppMeasurement.min.js`

## Adobe Demdex

`dpm.demdex.net`

## Facebook Pixel

`fbevents.js`

## Outbrain

`obtp.js`

## DoubleClick

`ad.doubleclick.net`

## Recommendation

Core policy renewal functionality must not depend on analytics or tracking services. Failure of marketing integrations must never interrupt customer transactions.

Note : These failures were observed in the console; their direct contribution to the renewal failure could not be confirmed.



# UI Architecture Concerns

## Evidence

- Two modal windows visible simultaneously
- Blank modal content
- Positioning differs across browsers
- User interaction completely blocked

## Impact

User is completely unable to dismiss the dialog or proceed with any renewal action.

## Potential Modal Duplication

showModal() called twice without lifecycle guard

## Potential Race Condition

API Error triggers multiple modal creation paths simultaneously

## Potential Layering (Z-Index) Issue

Overlapping stacking contexts cause rendering confusion



# Recommendations

Investigate: "smartech is not defined" error

Investigate: "Cannot read properties of undefined (reading '0')"

Review modal rendering framework and lifecycle management

Ensure failed third-party integrations cannot break renewal flow

Validate renewal flow using the specific policy/vehicle combination that reproduced the issue.

## Expected Outcomes

No overlapping dialogs

Meaningful error messages displayed

Successful premium retrieval

Successful policy renewal completion

Resilient behavior when optional integrations fail